

Chemistry 101 Laboratory Manual

Experiment 3: Acid-Base Titration

■■ SAFETY WARNING

This experiment involves the use of strong acids and bases. Always wear safety goggles, lab coat, and gloves. Work in a well-ventilated area or fume hood.

Objectives

1. Determine the concentration of an unknown acid solution
2. Practice proper titration technique
3. Calculate molarity from titration data

Materials

Equipment	Quantity	Chemicals	Amount
Burette	1	0.1 M NaOH	50 mL
Erlenmeyer flask	3	Unknown HCl	25 mL each
Pipette (25 mL)	1	Phenolphthalein	3 drops
Burette clamp	1	Distilled water	As needed

Procedure

Step 1: Prepare the burette

Rinse the burette with distilled water, then with small amounts of the NaOH solution. Fill the burette with 0.1 M NaOH solution. Record the initial volume.

Step 2: Prepare the flask

Use a pipette to transfer exactly 25.0 mL of unknown HCl solution into an Erlenmeyer flask. Add 2-3 drops of phenolphthalein indicator.

Step 3: Perform titration

Slowly add NaOH from the burette while swirling the flask constantly. As you approach the endpoint, add drops more slowly. Stop when the solution turns pale pink and remains that color for 30 seconds. Record the final volume.

Step 4: Repeat

Repeat the titration two more times for a total of three trials. Average the volumes used.

Data Collection

Trial	Initial Volume (mL)	Final Volume (mL)	Volume Used (mL)
1			
2			
3			
Average			

Calculations

Use the formula: $M_{\text{HCl}}V_{\text{HCl}} = M_{\text{NaOH}}V_{\text{NaOH}}$

Where M_{HCl} = molarity of HCl (unknown), V_{HCl} = 25.0 mL, M_{NaOH} = 0.1 M (NaOH), V_{NaOH} = average volume used

Post-Lab Questions

1. Why is it important to rinse the burette with NaOH solution before filling?
2. What would happen if you added too much NaOH past the endpoint?
3. Calculate the percent error if the actual concentration was 0.105 M.